AWS API Gateway – Execution Task

Objective

To create an **HTTP API** using AWS API Gateway and integrate it with AWS Lambda to expose a backend service over HTTP.

What is API Gateway?

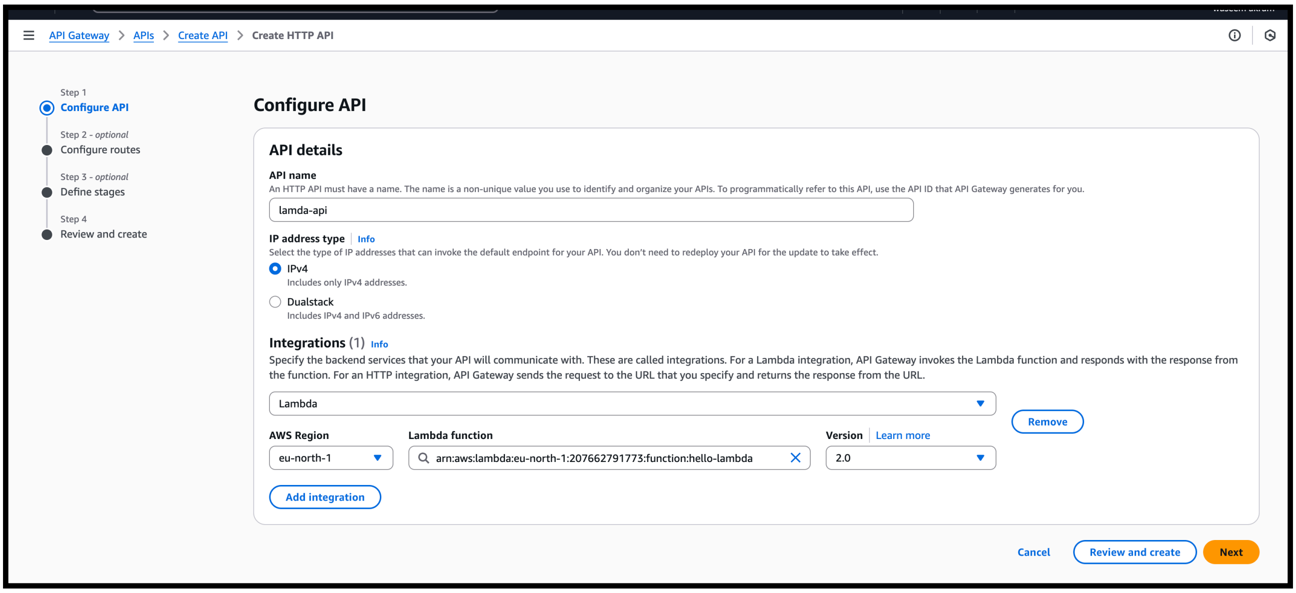
AWS API Gateway is a managed service that allows users to create, publish, and manage APIs. It acts as a front door for applications to access backend services such as AWS Lambda using HTTP requests.

Use cases :~ API GATEWAY AND LAMBDA configuration

Steps:

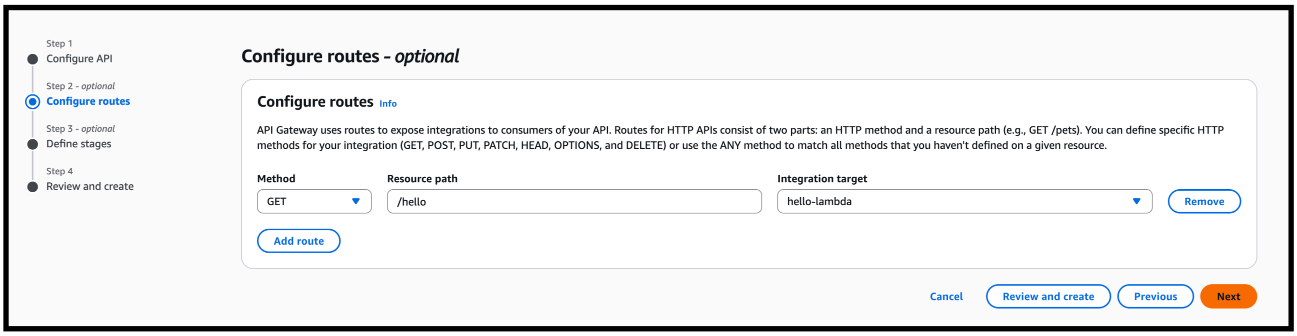
Created HTTP API

- API type: **HTTP**
- API name: `lamda-api`
- IP type: IPv4



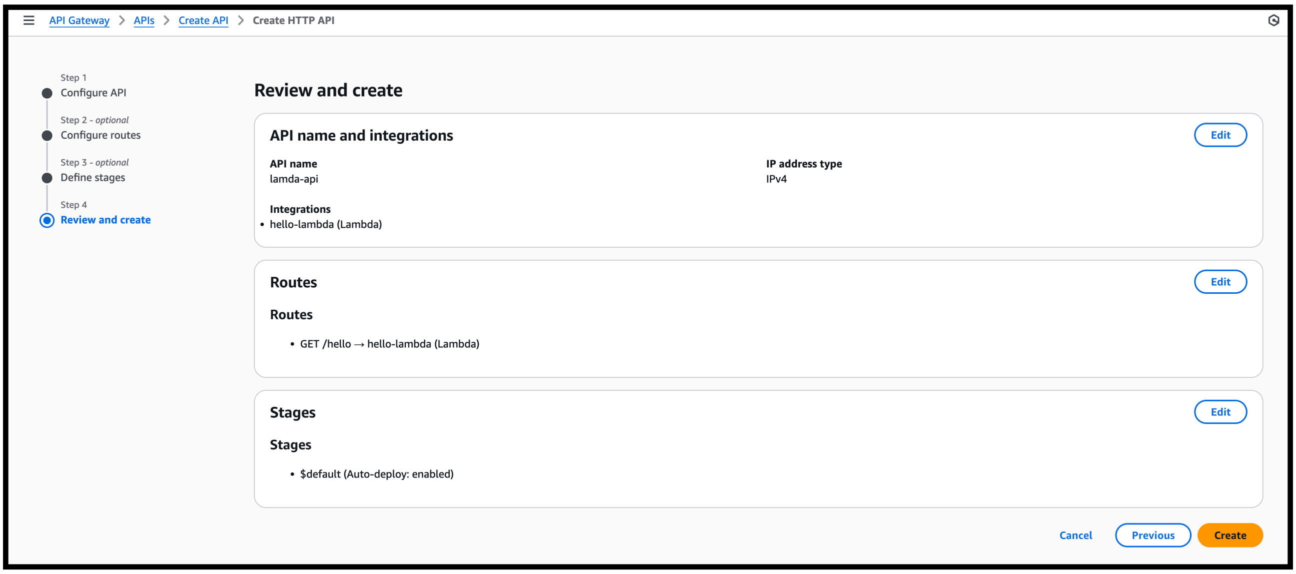
Integrated with Lambda

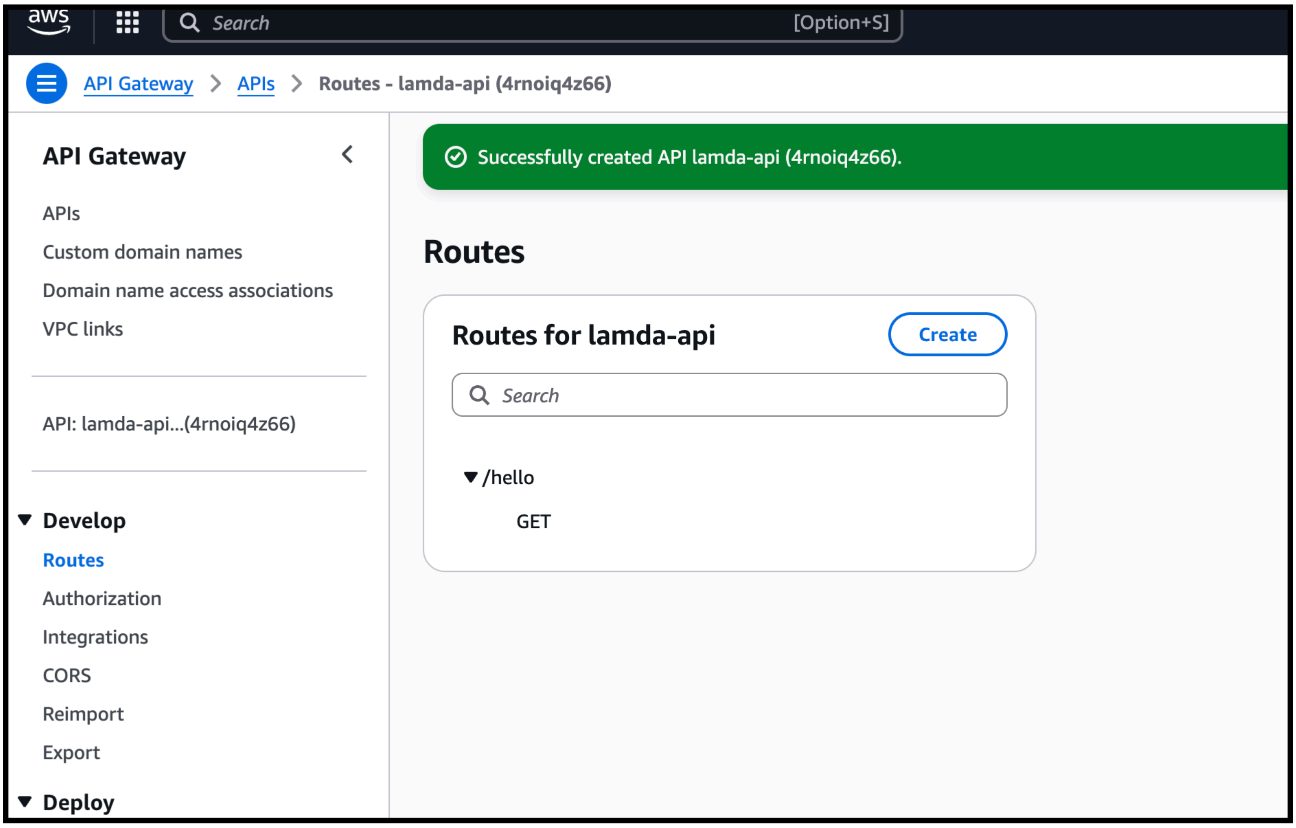
- Lambda function: `hello-lambda`
- Proper permission added automatically
- API Gateway can invoke Lambda



Created Route

- Route: `GET /hello`
- Method: GET
- Resource path working correctly





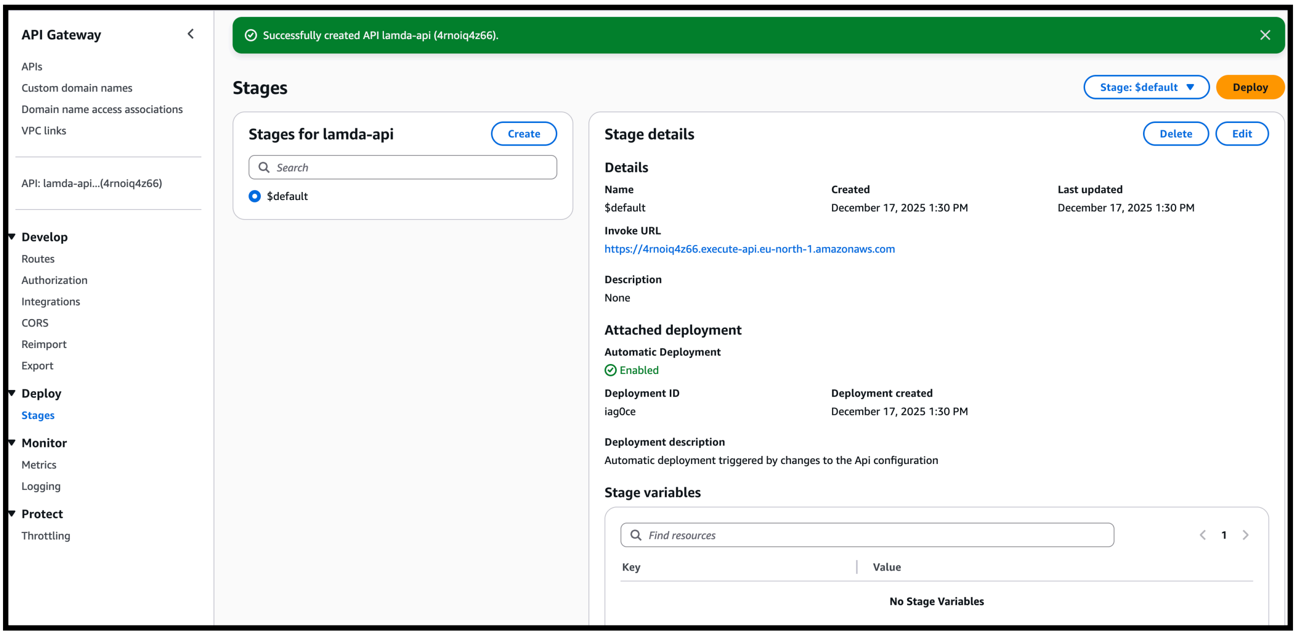
Stage Configured

- Stage: `$default`
- Auto-deploy enabled

Tested Successfully

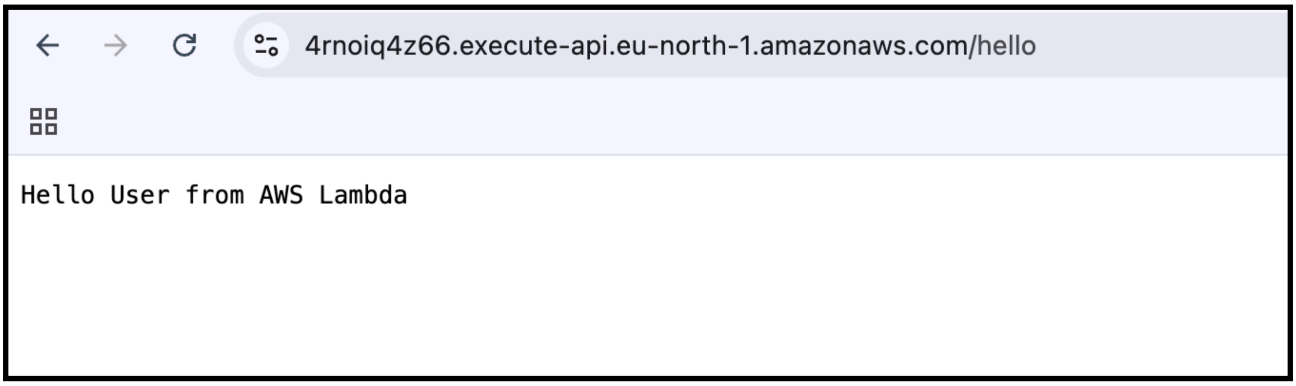
- Public endpoint opened in browser
- Output received:

```
Hello User from AWS Lambda
This confirms end-to-end working.
```



—> copy the invoke url and paste in browser with /hello

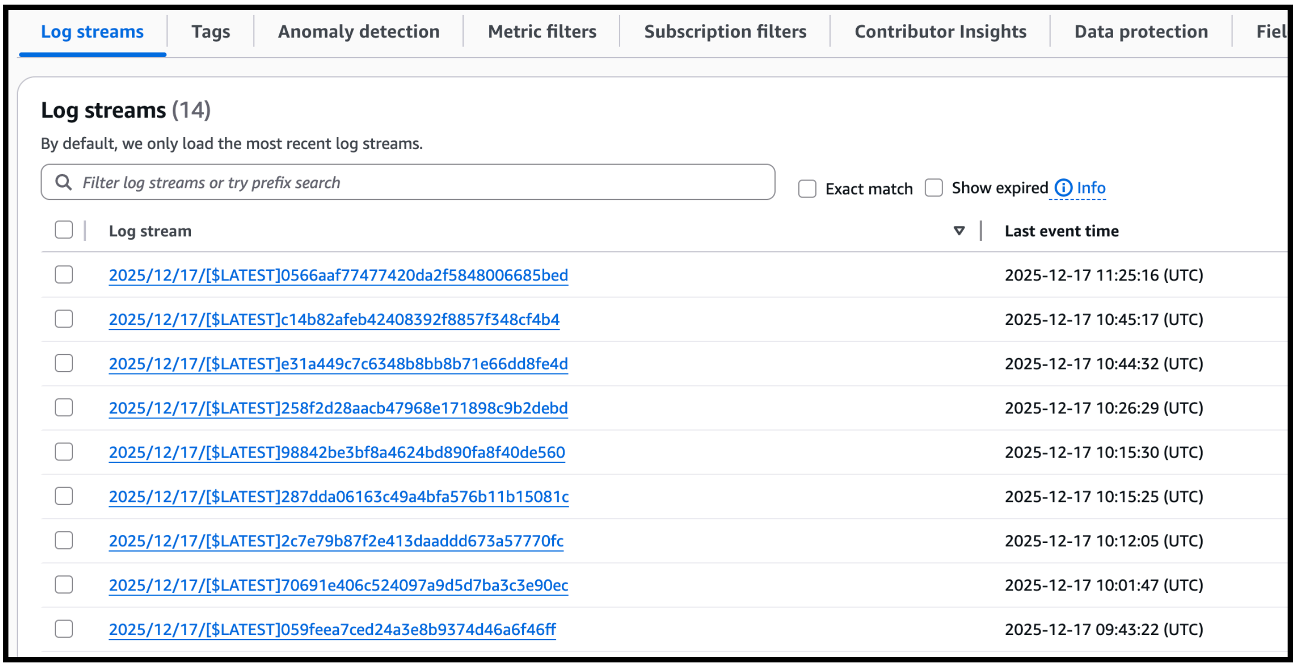
—> API Endpoint Testing



- `This confirms end-to-end working.`

Monitoring with CloudWatch

- API Gateway automatically sends execution logs to CloudWatch
- Lambda execution logs are captured separately
- Helps in monitoring request and response flow



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Architecture Flow

```
Client (Browser)
     ↓
API Gateway (HTTP API)
     ↓
AWS Lambda
     ↓
Response back to Client
```

Key Observations

- API Gateway routes HTTP requests to Lambda
- No server provisioning required
- Supports automatic scaling
- Tight integration with AWS Lambda